

Aditya Kumar Sethia

Ambitious third-year B.Tech student with strong foundational computer science knowledge, combining generative modeling and high-performance computing experience to architect scalable, reliable, and efficient software solutions within a cooperative engineering environment.

+91 9063975622 | adityakumarsethia@gmail.com | github.com/AdityaKumarSethia | linkedin.com/in/aditya-kumar-sethia

TECHNICAL SKILLS

Languages: Python, C/C++, Bash, SQL, Java

Data Structures & Algorithms: Strong command of core CS fundamentals, built through extensive practice with algorithmic problems spanning arrays, trees, graphs, and dynamic programming

Frameworks: PyTorch, PyTorch Lightning, TensorFlow, LangChain

Tools & Systems: Git/GitHub, FastAPI, Linux (Ubuntu), Flask, Streamlit

Core Domains: Deep Learning, Machine Learning, Generative AI, Data Science, Probabilistic Modelling, Parallel Computing

Soft Skills: Team Player, Time Management, Critical Thinking, Communication

EXPERIENCE

Open-Source Contributor (PyTorch Ecosystem)

Early 2026 – April 2026

neural-lam & Kornia

- Refactored neural-lam in Python, an object-oriented language, applying strict Object-Oriented Programming (OOP) paradigms, S.O.L.I.D. principles, and modular design to enable scalable, reliable probabilistic forecasting.
- Authored mllam PR #193, validated through rigorous testing, to resolve critical DDP initialization bugs, stabilizing PyTorch Lightning execution and improving overall framework reliability.
- Drove innovative optimizations to linear color transformations within the Kornia computer vision library, merging multiple core pull requests to accelerate processing efficiency and performance.

Software Engineering Intern (Python & AI)

June 2025 – July 2025

Eficens Software Services

- Engineered scalable software solutions — including a Retrieval-Augmented Generation (RAG) application with a secure FastAPI backend — leveraging LangChain, PyTorch, and ChromaDB for web content vectorization and LLM-based querying, while collaborating with Technical Program Managers and Product Managers to define requirements and uphold product quality, reliability, and performance.
- Collaborated within a cooperative team environment to author a comprehensive suite of 300+ Bash scripts automating Ubuntu 24.04 security audits, improving operational efficiency, consistency, and accountability while enforcing strict compliance with industry-standard CIS benchmarks.

PROJECTS

Generative Physics Simulation (Diffusion)

- Developed a PyTorch-based latent diffusion model — a cutting-edge technology in generative AI — to simulate complex fluid dynamics and chemical interactions by directly solving Partial Differential Equations (PDEs).
- Built an interactive JS/HTML visualization dashboard to render and analyze real-time inferences of the Shallow Water Equations, improving observability and performance monitoring.

Plant Disease Detection (Computer Vision)

- Trained and optimized a robust deep learning classifier in TensorFlow, engineering technical solutions to accurately identify multiclass crop diseases from raw agricultural imagery.
- Engineered efficient data augmentation and image preprocessing pipelines to significantly improve model generalization, reliability, and inference performance under varying visual conditions.

EDUCATION

Bachelor of Technology (B.Tech) in Artificial Intelligence & Data Science

Expected 2027

IcfaiTech, IFHE Hyderabad | CGPA: 9.77